

Task 3

Title

Student Data Analysis using Pandas

Objective

To analyze student performance using Python Pandas library.

Tools Used

- Python
- Pandas
- CSV File Handling

Key Operations

- Data loading from CSV
- Missing value handling
- Filtering and grouping data
- Calculating averages
- Generating insights

Screenshot

- Screenshot1

The screenshot shows a Visual Studio Code editor window with a dark theme. The Explorer sidebar on the left displays a project structure for 'PYTHON_INTERNSHIP_TASKS', including folders for 'Screenshots', 'Task1_File_Handling', 'Task2_API_Project', and 'Task3_Pandas_Project'. Under 'Task3_Pandas_Project', there is an 'Output' folder containing 'processed_students...', 'analysis.py', and 'students.csv'. The main editor area shows the 'analysis.py' file, which contains a Python script for data analysis using pandas. The script includes comments and code for loading a dataset, checking for missing values, cleaning data, calculating average scores, filtering top students, grouping by subject, and identifying high attendance students. The status bar at the bottom indicates the current position is Line 53, Column 5, with 4 spaces, UTF-8 encoding, CRLF line endings, and Python 3.14 (64-bit) interpreter.

```
7
8
9 try:
10     # Load dataset
11     df = pd.read_csv("students.csv")
12
13     print("\n===== ORIGINAL DATA =====")
14     print(df)
15
16     # Check missing values
17     print("\n===== MISSING VALUES CHECK =====")
18     print(df.isnull().sum())
19
20     # Data Cleaning (if any missing values)
21     df = df.fillna(0)
22
23     # Calculate Average Score
24     df["Average"] = df[["Math", "Science", "English"]].mean(axis=1)
25
26     # Filter: Students with average > 70
27     top_students = df[df["Average"] > 70]
28
29     # Grouping (basic insight)
30     avg_subject_scores = df[["Math", "Science", "English"]].mean()
31
32     print("\n===== TOP STUDENTS (Average > 70) =====")
33     print(top_students)
34
35     print("\n===== SUBJECT-WISE AVERAGE =====")
36     print(avg_subject_scores)
37
38     # Attendance insight
39     high_attendance = df[df["Attendance"] >= 90]
40
41     print("\n===== HIGH ATTENDANCE STUDENTS =====")
42     print(high_attendance)
43
44     # Save processed data
```

- Screenshot2

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\kumar\OneDrive\Desktop\Python_Internship_Tasks\Task3_Pandas_Project> python analysis.py

===== ORIGINAL DATA =====
   Name  Math  Science  English  Attendance
0  Aman    78     85     72         90
1  Riya    88     92     81         95
2  Rahul   45     56     60         70
3  Sneha   91     89     95         98
4  Vikas   66     70     72         85
5  Priya   82     76     88         92
6  Arjun   55     60     58         65
7  Neha    95     98     96         99
8  Karan   40     50     45         60
9  Simran  73     77     80         88

===== MISSING VALUES CHECK =====
Name      0
Math      0
Science   0
English   0
Attendance 0
dtype: int64

===== TOP STUDENTS (Average > 70) =====
   Name  Math  Science  English  Attendance  Average
0  Aman    78     85     72         90  78.333333
1  Riya    88     92     81         95  87.000000
3  Sneha   91     89     95         98  91.666667
5  Priya   82     76     88         92  82.000000
7  Neha    95     98     96         99  96.333333
9  Simran  73     77     80         88  76.666667

===== SUBJECT-WISE AVERAGE =====
Math      71.3
Science   75.3
English   74.7
dtype: float64

===== HIGH ATTENDANCE STUDENTS =====
   Name  Math  Science  English  Attendance  Average
0  Aman    78     85     72         90  78.333333

===== SUBJECT-WISE AVERAGE =====
Math      71.3
Science   75.3
English   74.7
dtype: float64

===== HIGH ATTENDANCE STUDENTS =====
   Name  Math  Science  English  Attendance  Average
0  Aman    78     85     72         90  78.333333
1  Riya    88     92     81         95  87.000000
3  Sneha   91     89     95         98  91.666667
5  Priya   82     76     88         92  82.000000
7  Neha    95     98     96         99  96.333333

Processed data saved to processed_students.csv
PS C:\Users\kumar\OneDrive\Desktop\Python_Internship_Tasks\Task3_Pandas_Project>
```

- Screenshot3

```
analysis.py | processed_students.csv X | students.csv
Task3_Pandas_Project > Output > processed_students.csv > data
1  Name,Math,Science,English,Attendance,Average
2  Aman,78,85,72,90,78.33333333333333
3  Riya,88,92,81,95,87.0
4  Rahul,45,56,60,70,53.666666666666664
5  Sneha,91,89,95,98,91.66666666666667
6  Vikas,66,70,72,85,69.33333333333333
7  Priya,82,76,88,92,82.0
8  Arjun,55,60,58,65,57.666666666666664
9  Neha,95,98,96,99,96.33333333333333
10 Karan,40,50,45,60,45.0
11 Simran,73,77,80,88,76.66666666666667
12
```

Insights

- Top performers identified based on average marks
- Subject-wise performance analyzed
- Attendance impact observed

Outcome

Successfully analyzed dataset and generated meaningful insights using Pandas.